

BODO, NorwayWMO: **011520**Lat: **67.267N**Lon: **14.359E**Elev: **13**StdP: **101.17**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.4	-9.0	-16.4	0.9	-9.1	-14.9	1.0	-8.0	19.3	0.6	17.6	0.5	8.8	90	0.823

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	22.5	15.7	20.5	14.8	18.7	13.8	16.4	21.1	15.3	19.4	14.3	17.7	4.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.4	10.3	17.8	13.5	9.7	16.7	12.7	9.2	15.7	46.1	21.1	43.0	19.4	40.2	17.7	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.3	12.8	DB	-12.0	25.7	1.8	2.2	-13.3	27.3	-14.4	28.6	-15.4	29.8	-16.7	31.3	(4)
(5)			WB	-12.8	18.0	1.6	1.4	-14.0	19.0	-14.9	19.8	-15.7	20.6	-16.9	21.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-0.8	-1.2	0.1	3.4	7.3	10.6	13.6	13.2	10.3	5.8	2.4	0.6	(6)	
	DBStd	6.24	3.96	4.07	3.44	3.06	3.26	3.13	3.09	2.74	2.87	3.28	3.56	3.95	(7)	
	HDD10.0	1968	334	313	308	198	97	28	2	2	31	134	229	292	(8)	
	HDD18.3	4697	592	546	566	447	341	233	152	160	241	389	479	550	(9)	
	CDD10.0	321	0	0	0	1	15	46	113	103	39	4	0	0	(10)	
	CDD18.3	8	0	0	0	0	0	1	5	2	0	0	0	0	(11)	
	CDH23.3	39	0	0	0	0	0	2	31	7	0	0	0	0	(12)	
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.4	8.4	8.2	7.3	5.9	5.4	4.7	4.1	4.5	5.7	6.8	7.5	7.9	(14)	
(15) Precipitation	PrecAvg	1059	90	70	73	61	54	60	81	87	123	143	107	110	(15)	
	PrecMax	1528	247	168	194	156	128	137	191	238	275	283	241	260	(16)	
	PrecMin	670	0	3	10	16	9	11	11	16	19	31	1	4	(17)	
	PrecStd	213	53	32	39	32	30	29	44	55	65	62	54	56	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	7.1	8.1	14.1	20.4	23.0	26.4	24.1	19.1	14.6	10.5	8.6	(19)	
		MCWB	5.9	5.5	4.7	8.0	12.6	15.3	18.0	16.7	13.9	10.4	8.3	6.9	(20)	
	2%	DB	6.5	6.0	6.5	11.1	17.0	20.2	23.5	21.6	17.1	12.8	9.1	7.6	(21)	
		MCWB	4.8	4.6	4.5	6.4	10.8	13.6	16.3	15.8	12.6	9.2	7.3	6.0	(22)	
	5%	DB	5.5	5.1	5.6	9.2	14.6	18.0	21.0	19.6	15.7	11.2	8.2	6.7	(23)	
		MCWB	3.9	3.8	3.8	5.4	9.7	12.4	15.3	14.8	11.9	8.7	6.1	5.2	(24)	
	10%	DB	4.3	4.2	4.8	7.5	12.3	16.0	19.0	17.8	14.2	10.1	7.2	5.8	(25)	
		MCWB	2.8	2.7	3.1	4.7	8.1	11.6	14.4	13.7	11.3	8.1	5.1	4.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	6.1	6.1	8.4	13.0	15.7	19.0	17.6	14.6	11.0	8.9	7.8	(27)	
		MCDB	7.0	6.5	6.8	13.6	18.8	21.6	24.8	22.7	18.1	13.2	9.8	8.2	(28)	
	2%	WB	5.2	5.0	5.1	6.9	11.3	14.0	16.9	16.2	13.3	10.0	7.7	6.5	(29)	
		MCDB	5.9	5.7	6.1	9.7	16.3	19.1	21.8	20.5	15.7	11.9	8.8	7.1	(30)	
	5%	WB	4.0	3.9	4.1	6.1	9.9	12.9	15.8	15.2	12.6	9.2	6.5	5.3	(31)	
		MCDB	5.1	4.8	5.3	8.1	13.9	17.1	20.2	18.7	14.7	10.8	7.8	6.4	(32)	
	10%	WB	2.9	2.8	3.2	5.2	8.6	11.9	14.7	14.2	11.7	8.3	5.3	4.2	(33)	
		MCDB	4.2	4.0	4.5	7.1	11.7	15.1	18.3	17.1	13.8	9.9	6.8	5.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	3.9	4.1	4.5	5.0	4.9	5.3	5.2	4.5	3.6	3.5	3.8	(35)	
		MCDBR	4.5	4.3	4.5	6.3	8.3	8.9	9.5	8.7	6.3	4.2	4.0	4.3	(36)	
	5% WB	MCWBR	4.2	4.2	3.6	3.8	4.2	4.4	4.4	4.2	3.4	3.1	3.5	4.2	(37)	
		MCWBR	4.5	4.3	4.3	5.1	7.7	7.8	8.7	7.8	5.4	3.9	4.2	4.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.365	0.280	0.313	0.334	0.340	0.339	0.347	0.336	0.320	0.289	0.349	N/A	(40)	
		taud	2.639	2.176	2.272	2.352	2.427	2.486	2.505	2.526	2.539	2.312	2.596	N/A	(41)	
	Ebn at Noon		176	540	731	822	858	868	848	821	744	532	157	0	(42)	
		Edh at Noon	16	48	76	91	95	92	88	77	60	42	14	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.04	0.37	1.50	3.19	4.49	4.92	4.50	3.36	1.78	0.63	0.10	0.00	(44)	
	RadStd		0.01	0.07	0.22	0.31	0.32	0.60	0.52	0.37	0.20	0.12	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.34	N/A	N/A	N/A	N/A	N/A	-93	-133	N/A	N/A
(47) Regional (24 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page